RAG-Powered Telecom Customer Service Assistant

Author: Janhavi Ghuge

Project Overview

This project builds a production-style **Retrieval-Augmented Generation (RAG)** system for a telecom company's customer service team. The system answers complex, policy-specific questions by grounding every response in a 22-document internal knowledge base — eliminating hallucinations and keeping answers consistent across agents.

Knowledge Base Structure

The knowledge base consists of 22 Markdown files organised across four folders:

- `customer-service` — complaint handling, escalation paths, SLA compensation
- `policies` — refund rules, early termination fees, uptime guarantees
- `procedures` — step-by-step cancellation, equipment return, provisioning
- `products` — plans, pricing, bundles, add-ons

What This Notebook Covers

1. **RAG Architecture** — how each pipeline component works in the telecom context
2. **Pipeline Development** — four progressively improved RAG versions with hyperparameter tuning
3. **System Evaluation** — six real-world test prompts ranging from medium to very hard difficulty

Part 1: Understanding RAG Architecture

I am building a pipeline where each stage has one job and hands off to the next.

The Document Loader reads all 22 markdown files from the telecom knowledge base and tags each one with the folder it came from (customer-service, policies, procedures, products) as metadata.

The Text Splitter breaks each document into smaller overlapping chunks so each piece represents one focused idea. For example one chunk might cover only the Fiber 250 SLA uptime guarantee instead of mixing it with cancellation policies.

The Embedding Model converts every chunk into a vector of numbers that captures its meaning. So a chunk about billing disputes and a chunk about refund timelines will have vectors close together even if the exact words are different.

The Vector Store (ChromaDB) saves all these vectors so that when a customer service agent asks a question, it can quickly find the chunks whose meaning is closest to that question across all 22 documents.

The Retriever sits on top of the vector store and fetches the top k most similar chunks for a given query.

The LLM (gpt-4-turbo) receives the agent's question plus the retrieved telecom document chunks as context and generates an accurate answer grounded in our actual policies and product specs rather than making things up from general training data.

Understanding why would a standard LLM (without RAG) be insufficient for this task?

The first reason is that the LLM has no knowledge of our company's specific products, pricing, or SLA terms. For example our Fiber 250 plan has a specific uptime guarantee and a specific credit amount for violations. The LLM has never seen these internal documents so it would either refuse to answer or confidently hallucinate a wrong number, which creates real legal and financial risk.

The second reason is that our documents change over time. Prices get updated, new plans launch, and SLA thresholds get revised. The LLM has a training cutoff and knows nothing about changes after that date. RAG solves this because we just update the markdown files and the retriever automatically surfaces the latest information with zero retraining.

The third reason is that our knowledge base has 22 documents across 4 categories and a plain LLM cannot distinguish which specific procedure applies to which situation. This is because the LLM has no awareness that these categories even exist in our system. It was trained on general internet text so when asked about a billing dispute it might blend together generic customer service advice from multiple sources instead of going directly to our billing-dispute-resolution.md file. RAG fixes this because the retriever searches the actual documents and pulls the exact section that matches the query, so the LLM is forced to answer from the right source.

The fourth reason is consistency. A plain LLM might answer the same question differently each time because it is generating from probability distributions over its training data. Two agents asking the same cancellation question could get two different answers. RAG ensures every agent gets the answer pulled from the same source document every single time, making responses consistent and auditable.

Understanding why hyperparameter tuning matters in a RAG system.

Chunk Size: When we split documents into chunks, the size of each chunk directly affects what the retriever finds. If chunks are too small, a question about SLA compensation might retrieve the uptime percentage but miss the compensation clause because it landed in a different tiny chunk that never got retrieved. If chunks are too large, one chunk mixes five plan tiers together and the retriever cannot tell which tier actually matches the query.

k (number of retrieved documents): k controls how many chunks the retriever fetches before passing context to the LLM. The whole point of k is to give the LLM enough relevant information to answer the question completely. Now in our telecom knowledge base, one customer question can require information from multiple documents at the same time. For example if a business customer asks about an SLA violation, a complete answer needs the uptime threshold from service-level-agreements.md AND the escalation steps from escalation-procedures.md AND the compensation amount from a third document. These are three separate files.

If k is set to 2, the retriever only fetches 2 chunks total. There is a very high chance both chunks come from the same document, meaning the other two documents never even get seen by the LLM. The agent then gets a half answer that covers the uptime threshold but says nothing about compensation or escalation. The customer walks away with incomplete information.

If k is set too high like 50, the LLM gets flooded with chunks from entertainment services and international calling plans that have nothing to do with the SLA question, which confuses the answer.

Reranking: Without reranking, similarity search ranks chunks purely by how close their embedding vectors are to the query vector. This is not the same as actual relevance. A query about business SLA compensation might surface a home fiber troubleshooting guide just because the word internet appears many times in both. Cohere Rerank fixes this by reading each chunk alongside the actual query and scoring for true relevance rather than just vector distance.

Part 2: Building the RAG Pipeline

Environment Setup

```
from google.colab import userdata
OPENAI_API_KEY = userdata.get('OPENAI_API_KEY')

from google.colab import drive
drive.mount('/content/drive')
import glob
import gradio as gr
from openai import OpenAI
```

```
Mounted at /content/drive
```

Set Environment Variable

```
import os
os.environ['OPENAI_API_KEY'] = OPENAI_API_KEY
```

Initialise Model

```
model = "gpt-4-turbo"
client = OpenAI()
```

Install Requirements

```
!pip install -r requirements_new.txt -q
```

```
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```

```
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ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behav
google-colab 1.0.0 requires requests==2.32.4, but you have requests 2.32.5 which is incompatible.
opentelemetry-exporter-gcp-logging 1.11.0a0 requires opentelemetry-sdk<1.39.0,>=1.35.0, but you have opentelemetry-sd
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-exporter-otlp-proto-common==1.38.0, but you have
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-proto==1.38.0, but you have opentelemetry-proto
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-sdk~=1.38.0, but you have opentelemetry-sdk 1.40
google-adk 1.26.0 requires opentelemetry-api<1.39.0,>=1.36.0, but you have opentelemetry-api 1.40.0 which is incompat
google-adk 1.26.0 requires opentelemetry-sdk<1.39.0,>=1.36.0, but you have opentelemetry-sdk 1.40.0 which is incompat
```

Import LangChain Dependencies

```
from langchain_community.document_loaders import DirectoryLoader, TextLoader
from langchain_text_splitters import RecursiveCharacterTextSplitter
from langchain_core.documents import Document
from langchain_openai import OpenAIEmbeddings, ChatOpenAI
from langchain_community.vectorstores import Chroma
```

Load all 22 markdown files

This loops through each subfolder (customer-service, policies, procedures, products) and tags every document with the folder name as doc_type metadata.

```
folders = glob.glob("/content/drive/MyDrive/Customer-knowledge-base/*")

documents = []
for folder in folders:
  doc_type = os.path.basename(folder)
  loader = DirectoryLoader(folder, glob="**/*.md", loader_cls=TextLoader)
  folder_docs = loader.load()
  for doc in folder_docs:
    doc.metadata["doc_type"] = doc_type
    documents.append(doc)
```

```
len(documents)
```

```
22
```

Chunking the data using text_splitters

I am chunking the documents because a full document covers many topics at once, making it a poor match for any specific query. Smaller chunks each represent one focused idea, so the retriever can pinpoint the exact section that answers the question.

I chose chunk_size of 1000 because I looked at the knowledge base and noticed that most meaningful sections like one SLA tier, one refund rule, one troubleshooting procedure are roughly 800 to 1200 characters long. So 1000 felt like a reasonable starting point that would capture one complete idea without mixing two together. The chunk_overlap of 200 makes sure nothing important gets cut off at a chunk boundary.

```
def chunk_data(data, chunk_size=1000):
  text_splitter = RecursiveCharacterTextSplitter(chunk_size=chunk_size, chunk_overlap=200)
  chunk = text_splitter.split_documents(data)
  return chunk
```

```
chunks = chunk_data(documents)
print(f'this document has {len(chunks)} chunks in the data')
```

```
this document has 205 chunks in the data
```

Building the Vector Store

The chunks are still raw text and computers cannot understand meaning in text directly. OpenAIEmbeddings converts each chunk into 1536 numbers called a vector where similar meaning chunks end up close together in space.

ChromaDB then stores all these vectors so when a query comes in it can instantly find the most relevant chunks by finding the closest vectors. ChromaDB is a vector database which works differently than the normal database, instead of exact matching it searches by similarity. When a query comes in it converts that query into a vector and then search which stored vectors are closest to this query vector.

```
!pip install -q chromadb
```

```
embeddings = OpenAIEmbeddings()
db_name = "vector_db"

if os.path.exists(db_name):
  vector_store = Chroma(persist_directory=db_name, embedding_function=embeddings)
  vector_store.delete_collection()
else:
  vector_store = Chroma(persist_directory=db_name, embedding_function=embeddings)

vector_store = Chroma.from_documents(chunks, embeddings, persist_directory=db_name)
print(f'this vector database has {vector_store._collection.count()} chunks')

sample_embedding = vector_store._collection.get(limit=1, include=['embeddings'])['embeddings'][0]
print(len(sample_embedding))
```

```
/tmp/ipykernel_3983/784347885.py:8: LangChainDeprecationWarning: The class `Chroma` was deprecated in LangChain 0.2.9
  vector_store = Chroma(persist_directory=db_name, embedding_function=embeddings)
this vector database has 205 chunks
1536
```

Building the Retriever and Connecting it to the LLM

Now that my vectors are stored in ChromaDB I need something that can actually go and fetch the right chunks when a question comes in. That is what the retriever does. It takes the query, converts it into a vector, searches ChromaDB for the closest matching chunks and returns them. I then connect it to gpt-4-turbo through a prompt template that instructs the LLM to answer only from the retrieved context and not from its general training data.

I am using search_type as similarity which means it ranks chunks purely by how close their vectors are to the query vector.

I set k to 6 which means the retriever fetches the top 6 most similar chunks. I chose 6 because our telecom questions sometimes need information from more than one document, for example a billing question might need both the billing dispute file and the refund policy file. Fetching only 1 or 2 chunks risks missing that second document entirely. But fetching too many like 20 would flood the LLM with irrelevant chunks from unrelated documents like entertainment services or international calling plans.

I set temperature to 0 on the LLM which means it gives the most deterministic and factual answer possible with no creativity or randomness. This is important for a customer service system where we want consistent and accurate answers every time not creative ones.

The prompt template instructs the LLM to answer only from the retrieved context and not from its general training data, keeping it grounded in our actual telecom documents.

Full flow is like → retriever fetches chunks from ChromaDB → format_docs joins chunks into text → prompt template wraps everything → LLM generates answer → StrOutputParser extracts clean text

```
from langchain_core.prompts import ChatPromptTemplate, MessagesPlaceholder
from langchain_core.output_parsers import StrOutputParser
from langchain_core.runnables import RunnablePassthrough

llm = ChatOpenAI(model="gpt-4-turbo", temperature=0)
```

```
retriever = vector_store.as_retriever(search_type="similarity", search_kwargs={"k": 6})

prompt = ChatPromptTemplate([
    ("system", "Answer the user query based on the context: \n\n {context}"),
    ("user", "{input}"),
    MessagesPlaceholder("chat_history")
])

def format_docs(docs):
  return "\n\n".join(doc.page_content for doc in docs)

chain = (
    {"context": retriever | format_docs, "input": RunnablePassthrough(), "chat_history": lambda x: []}
    | prompt
    | llm
    | StrOutputParser()
)

response = chain.invoke("What are the SLA uptime guarantees for business customers?")
print(response)

def chat(message, history):
  response = chain.invoke(message)
  return response

gr.ChatInterface(chat).launch()
```

```
The SLA uptime guarantees for business customers are structured as follows:

1. **Standard Business SLA**:
   - **Uptime Guarantee**: 99.7% monthly availability

2. **Premium Business SLA**:
   - **Uptime Guarantee**: 99.9% monthly availability

3. **Enterprise SLA**:
   - **Uptime Guarantee**: 99.95% monthly availability

4. **Mission Critical SLA**:
   - **Uptime Guarantee**: 99.99% monthly availability

These guarantees reflect the commitment to maintaining service availability at different levels based on the tier of
/usr/local/lib/python3.12/dist-packages/gradio/chat_interface.py:347: UserWarning: The 'tuples' format for chatbot me
  self.chatbot = Chatbot(
It looks like you are running Gradio on a hosted Jupyter notebook, which requires `share=True`. Automatically setting

Colab notebook detected. To show errors in colab notebook, set debug=True in launch()
* Running on public URL: https://f64ee3d4cadc232364.gradio.live

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run `gradio deploy` from the terminal
```

- **Cloud Backup**: 100 OMR
- **VPN**: 250 OMR
- **Bundle Cost**: 500 OMR (if applicable, but here we consider individual services as they exceed bundle limits)

Grand Total: 750 + 1200 + 750 + 600 + 100 + 250 = 3650 OMR

This setup ensures that each store location is equipped with high-speed internet, all delivery drivers are connected, and the business's data and communications are secure and efficient.

Next steps: Deploy to Cloud Run

Version 2a — Reranking (Hyperparameter Tuning #1)

Hyperparameter Tuning 1 - Reranking the similarity search

The basic retriever I built works but it has one weakness. It ranks chunks purely by vector distance which is not the same as actual relevance. For example a query about business SLA compensation might surface a home fiber troubleshooting chunk just because the word internet appears many times in both.

To fix this I am adding CohereRerank. The idea is to first cast a wide net by fetching 20 chunks instead of 6, then let Cohere read each chunk alongside the actual query and score them for true relevance. Only the top 5 most relevant chunks survive and get passed to the LLM.

I set k to 20 because we want a large candidate pool for Cohere to work with. If we only gave Cohere 6 chunks the truly relevant ones might not even be in that pool. I set top_n to 5 because after reranking we only want the best ones going to the LLM, too many would add noise.

```
!pip install langchain-cohere -q
```

```
from langchain_core.prompts import ChatPromptTemplate, MessagesPlaceholder
from langchain_core.output_parsers import StrOutputParser
from langchain_core.runnables import RunnablePassthrough, RunnableLambda
from langchain_cohere import CohereRerank

os.environ["CO_API_KEY"] = userdata.get('CO_API_KEY')

llm = ChatOpenAI(model="gpt-4-turbo", temperature=0)
retriever = vector_store.as_retriever(search_type='similarity', search_kwargs={'k': 20})
reranker = CohereRerank(model="rerank-english-v3.0", top_n=5)

def retrieve_and_rerank(query):
  docs = retriever.invoke(query)
  return reranker.compress_documents(docs, query)

prompt = ChatPromptTemplate([
    ("system", "Answer the user query using the context provided: \n\n {context}"),
    ("user", "{input}"),
    MessagesPlaceholder("chat_history")
])

def format_docs(docs):
  return "\n\n".join(doc.page_content for doc in docs)

chain = (
    {"context": RunnableLambda(retrieve_and_rerank) | format_docs, "input": RunnablePassthrough(), "chat_history": l
    |prompt
    |llm
    |StrOutputParser()
)

response = chain.invoke("What are the SLA uptime guarantees for business customers?")
print(response)

def chat(message, history):
  response = chain.invoke(message)
  return response

gr.ChatInterface(chat).launch()
```

```
The SLA uptime guarantees for business customers are structured as follows:

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3. **Enterprise SLA**:
   - **Uptime Guarantee**: 99.95% monthly availability

4. **Mission Critical SLA**:
   - **Uptime Guarantee**: 99.99% monthly availability

These guarantees reflect the commitment to maintaining service availability at different levels based on the tier of
/usr/local/lib/python3.12/dist-packages/gradio/chat_interface.py:347: UserWarning: The 'tuples' format for chatbot me
  self.chatbot = Chatbot(
It looks like you are running Gradio on a hosted Jupyter notebook, which requires `share=True`. Automatically setting

Colab notebook detected. To show errors in colab notebook, set debug=True in launch()
* Running on public URL: https://d2a46094881a7aa60e.gradio.live

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run `gradio deploy` from the terminal
```

Chatbot

- **Monthly Cost for 50 Mailboxes**: 12 OMR x 50 = 600 OMR
- **SLA**: Standard email security SLA

Cloud Backup for Business Data

- **Product**: Custom Solution (not explicitly listed, assuming a typical cloud backup service)
- **Per-Unit Cost**: Custom pricing (assuming 100 OMR for medium business)
- **Total Monthly Cost**: 100 OMR
- **SLA**: Standard cloud service SLA

Site-to-Site VPN Connecting All Locations

- **Product**: Included in Business Fiber Pro features
- **Per-Unit Cost**: Included in Business Fiber Pro
- **Total Monthly Cost**: Included in Business Fiber Pro cost
- **SLA**: Included in Business Fiber Pro SLA

Bundle Pricing Consideration

Next steps: Deploy to Cloud Run

Version 2b — Query Expansion + Reranking (Hyperparameter Tuning #2)

The reranking I added works well but there is still one problem. The retriever only searches using the exact words the user typed. If the user asks "cancel contract" but our document says "early termination fee" the retriever might completely miss that document because the words do not match even though the meaning is identical.

Query expansion fixes this by asking the LLM to rephrase the original query 3 different ways before searching. So instead of searching once we search 4 times with 4 different phrasings, combine all the results, remove duplicates and then rerank the entire combined pool. This way vocabulary mismatches between how the agent asks and how the document is written no longer cause relevant chunks to be missed.

I set k to 10 per query because we are now running 4 searches so the total candidate pool before deduplication could be up to 40 chunks which is already a large enough pool for Cohere to rerank from.

```
from langchain_core.prompts import ChatPromptTemplate, MessagesPlaceholder
from langchain_core.output_parsers import StrOutputParser
from langchain_core.runnables import RunnablePassthrough, RunnableLambda
from langchain_cohere import CohereRerank

os.environ["CO_API_KEY"] = userdata.get('CO_API_KEY')
```

```
llm = ChatOpenAI(model="gpt-4-turbo", temperature=0)
retriever = vector_store.as_retriever(search_type='similarity', search_kwargs={'k': 10})
reranker = CohereRerank(model="rerank-english-v3.0", top_n=5)

def expand_query(query):
    """Use LLM to generate 3 query variations"""
    expansion_prompt = f"""Generate 3 different versions of this search query to get better search results.
Return only the 3 queries, one per line, no numbering.

Original query: {query}"""

    expandresponse = llm.invoke(expansion_prompt)
    queries = [q.strip() for q in expandresponse.content.strip().split('\n') if q.strip()]
    return [query] + queries[:3]  # Original + up to 3 expansions

def retrieve_with_expansion_and_rerank(query):
    # Expand query into multiple versions
    queries = expand_query(query)

    # Retrieve for each query and dedup ensures unique documents only
    all_docs = []
    seen = set()
    for q in queries:
        docs = retriever.invoke(q)
        for doc in docs:
            if doc.page_content not in seen:
                seen.add(doc.page_content)
                all_docs.append(doc)

    # Rerank all combined docs against original query
    return reranker.compress_documents(all_docs, query)

prompt = ChatPromptTemplate([
    ("system", "Answer the user query using the context provided: \n\n {context}"),
    ("user", "{input}"),
    MessagesPlaceholder("chat_history")
])

def format_docs(docs):
  return "\n\n".join(doc.page_content for doc in docs)

chain = (
    {"context": RunnableLambda(retrieve_with_expansion_and_rerank) | format_docs, "input": RunnablePassthrough(), "c
    |prompt
    |llm
    |StrOutputParser()
)

response = chain.invoke("What are the SLA uptime guarantees for business customers?")
print(response)

def chat(message, history):
  response = chain.invoke(message)
  return response

gr.ChatInterface(chat).launch()
```

```
The SLA uptime guarantees for business customers are structured as follows:

1. **Standard Business SLA**:
   - **Uptime Guarantee**: 99.7% monthly availability

2. **Premium Business SLA**:
   - **Uptime Guarantee**: 99.9% monthly availability

3. **Enterprise SLA**:
   - **Uptime Guarantee**: 99.95% monthly availability

4. **Mission Critical SLA**:
   - **Uptime Guarantee**: 99.99% monthly availability

These guarantees reflect the commitment to maintaining service availability at different levels based on the tier of
/usr/local/lib/python3.12/dist-packages/gradio/chat_interface.py:347: UserWarning: The 'tuples' format for chatbot me
  self.chatbot = Chatbot(
It looks like you are running Gradio on a hosted Jupyter notebook, which requires `share=True`. Automatically setting

Colab notebook detected. To show errors in colab notebook, set debug=True in launch()
* Running on public URL: https://2013c662617d24aade.gradio.live

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run `gradio deploy` from the terminal
```

Chatbot

at each location with guaranteed uptime, a mobile fleet plan for 30 delivery drivers with vehicle tracking, a VoIP phone system across all locations, POS connectivity at every store, email security for 50 employees, cloud backup for business data, and a site-to-site VPN connecting all locations. For each need, recommend the specific product or plan, state the per-unit and total monthly cost, identify the applicable SLA tier with its response time guarantees, and calculate the grand total monthly expenditure across all locations and services.

To set up a comprehensive telecom infrastructure for a medium-sized retail business with 5 store locations, we can recommend the following products and plans from the available options:

Dedicated Internet

- **Product**: Business Fiber Enterprise
- **Per-Unit Cost**: 275 OMR per location
- **Total Monthly Cost for 5 Locations**: 275 OMR x 5 = 1375 OMR

Next steps: Deploy to Cloud Run

Version 2c — Parent-Child Chunking + Query Expansion + Reranking (Hyperparameter Tuning #3)

Hyperparameter Tuning 3 - Parent Child Chunking

So far I have been searching and passing the same 1000 character chunks to the LLM. This creates a tension. Small chunks are good for retrieval because they represent one focused idea and match queries precisely. But small chunks are bad for the LLM because they lack surrounding context. For example a 1000 character chunk might contain the SLA compensation amount but cut off before explaining the conditions under which it applies.

Parent child chunking solves this tension by having two levels of chunks. Child chunks are small at 400 characters and are what gets embedded and searched. Parent chunks are large at 2000 characters and are what gets passed to the LLM. I keep a dictionary called child_to_parent that maps every child back to the parent it came from.

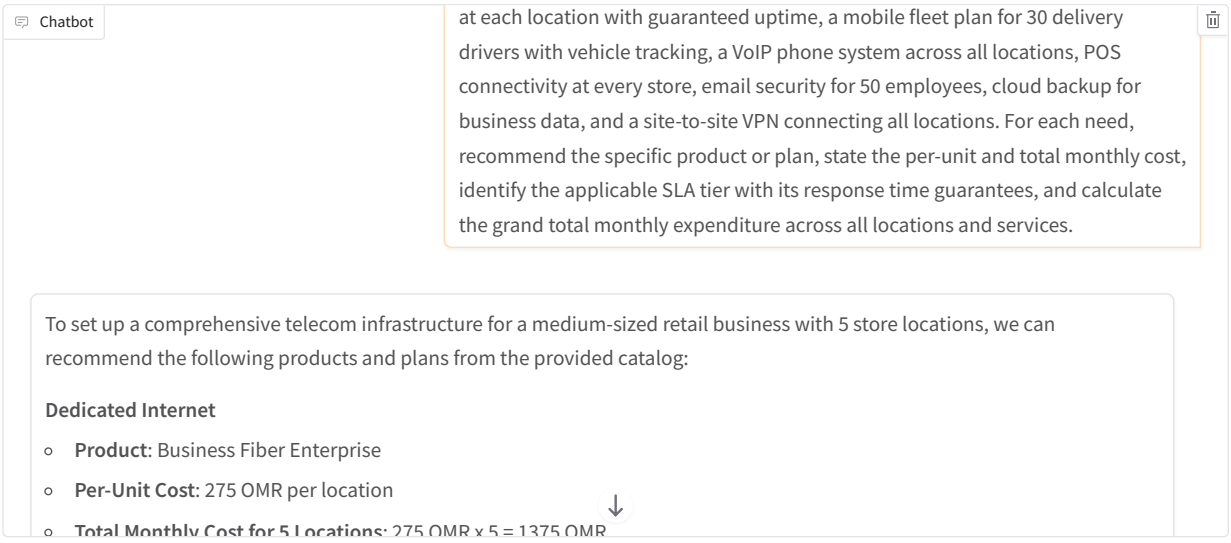
So during retrieval the child chunk finds the exact right section with precision. But before passing to the LLM I swap each child for its parent so the LLM gets the full surrounding context. The child finds the right page, the parent gives the LLM the whole chapter.

I set child chunk size to 400 because it is small enough to represent one specific idea for precise matching. I set parent chunk size to 2000 because it is large enough to contain the full context around that idea without mixing too many unrelated topics together.

```
from langchain_core.prompts import ChatPromptTemplate, MessagesPlaceholder
from langchain_core.output_parsers import StrOutputParser
from langchain_core.runnables import RunnablePassthrough, RunnableLambda
from langchain_cohere import CohereRerank

os.environ["CO_API_KEY"] = userdata.get('CO_API_KEY')

# ============================================================
# STEP 1: PARENT-CHILD CHUNKING (replaces original chunking)
# ============================================================

# PARENT = Large chunks (2000 chars) - what LLM will see
parent_splitter = RecursiveCharacterTextSplitter(chunk_size=2000, chunk_overlap=200)
parents = parent_splitter.split_documents(documents)

# CHILD = Small chunks (400 chars) - what we search on
child_splitter = RecursiveCharacterTextSplitter(chunk_size=400, chunk_overlap=50)

# Link each child to its parent
children = []
child_to_parent = {}

for parent in parents:
    parent_children = child_splitter.split_documents([parent])
    for child in parent_children:
        children.append(child)
        child_to_parent[child.page_content] = parent

# Build vector store on CHILDREN
embeddings = OpenAIEmbeddings()
vector_store = Chroma.from_documents(children, embeddings)

#Step 2....

llm = ChatOpenAI(model="gpt-4-turbo", temperature=0)
retriever = vector_store.as_retriever(search_type='similarity', search_kwargs={'k': 10})
reranker = CohereRerank(model="rerank-english-v3.0", top_n=5)

def expand_query(query):
    """Use LLM to generate 3 query variations"""
    expansion_prompt = f"""Generate 3 different versions of this search query to get better search results.
Return only the 3 queries, one per line, no numbering.

Original query: {query}"""

    expandresponse = llm.invoke(expansion_prompt)
    queries = [q.strip() for q in expandresponse.content.strip().split('\n') if q.strip()]
    return [query] + queries[:3]

def retrieve_with_expansion_and_rerank(query):
    # Expand query into multiple versions
    queries = expand_query(query)

    # Retrieve CHILDREN for each query
    all_children = []
    seen = set()
    for q in queries:
        docs = retriever.invoke(q)
        for doc in docs:
            if doc.page_content not in seen:
                seen.add(doc.page_content)
                all_children.append(doc)

    # Rerank children chunks
    reranked_children = reranker.compress_documents(all_children, query)

    # Swap children for PARENTS (more context for LLM)
    parents_out = []
    seen_parents = set()
    for child in reranked_children:
        parent = child_to_parent.get(child.page_content, child)
        if parent.page_content not in seen_parents:
            seen_parents.add(parent.page_content)
            parents_out.append(parent)

    return parents_out
```

```
prompt = ChatPromptTemplate([
    ("system", "Answer the user query using the context provided: \n\n {context}"),
    ("user", "{input}"),
    MessagesPlaceholder("chat_history")
])

def format_docs(docs):
  return "\n\n".join(doc.page_content for doc in docs)

chain = (
    {"context": RunnableLambda(retrieve_with_expansion_and_rerank) | format_docs, "input": RunnablePassthrough(), "c
    |prompt
    |llm
    |StrOutputParser()
)

response = chain.invoke("What are the SLA uptime guarantees for business customers?")
print(response)

def chat(message, history):
  response = chain.invoke(message)
  return response

gr.ChatInterface(chat).launch()
```

```
The SLA uptime guarantees for business customers are structured into different tiers, each designed to meet varying l

1. **Standard Business SLA**: 99.7% monthly availability
2. **Premium Business SLA**: 99.9% monthly availability
3. **Enterprise SLA**: 99.95% monthly availability
4. **Mission Critical SLA**: 99.99% monthly availability

Each tier offers a higher level of uptime guarantee to accommodate the criticality and operational demands of differe
/usr/local/lib/python3.12/dist-packages/gradio/chat_interface.py:347: UserWarning: The 'tuples' format for chatbot me
  self.chatbot = Chatbot(
It looks like you are running Gradio on a hosted Jupyter notebook, which requires `share=True`. Automatically setting

Colab notebook detected. To show errors in colab notebook, set debug=True in launch()
* Running on public URL: https://29f923cf9541c65f47.gradio.live

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run `gradio deploy` from the terminal
```



Next steps: Deploy to Cloud Run

Version 2d — Metadata Filtering + All Previous Strategies (Hyperparameter Tuning #4)

Hyperparameter Tuning 4 - Metadata Filtering

Even with everything I have built so far, ChromaDB still searches across all 22 documents for every single query. That means a question about a billing dispute has to compete against chunks from entertainment services, international calling plans and mobile packages before reranking can clean things up. This adds unnecessary noise to the candidate pool.

Metadata filtering solves this by narrowing the search space before similarity search even starts. Remember when I loaded the documents I tagged every chunk with the folder it came from as doc_type. Now I use that tag to restrict ChromaDB to only search inside the relevant folders.

I added a detect_filters function that sends the query to the LLM and asks it to classify which of our four categories are relevant. I am only filtering by doc_type and not by year or source because those fields were never stored as metadata during loading, so ChromaDB has nothing to filter on for them. For example a question about an SLA violation would be classified as policies and procedures. ChromaDB then completely skips the products and customer-service folders. The candidate pool going into reranking is already cleaner and more focused from the start which leads to better final answers.

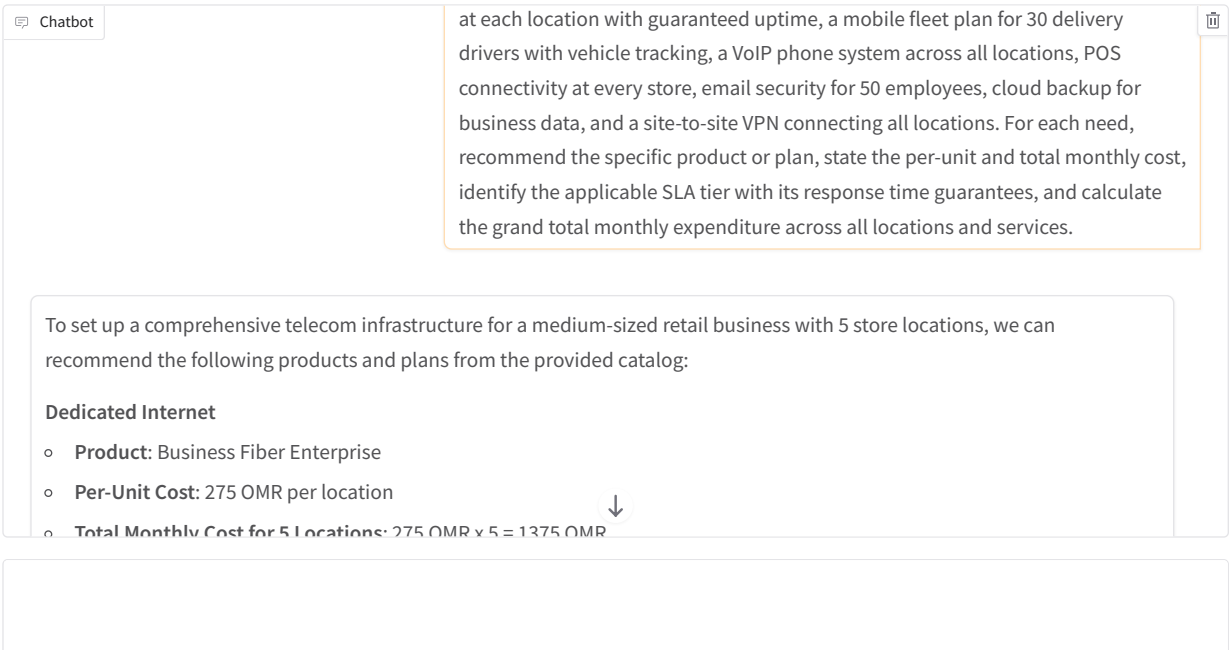
The complete pipeline for every query is now: detect relevant categories, filter ChromaDB to those categories, expand the query into 4 versions, retrieve child chunks, rerank, swap to parents, LLM generates the answer.

```
from langchain_core.prompts import ChatPromptTemplate, MessagesPlaceholder
from langchain_core.output_parsers import StrOutputParser
from langchain_core.runnables import RunnablePassthrough, RunnableLambda
from langchain_cohere import CohereRerank

os.environ["CO_API_KEY"] = userdata.get('CO_API_KEY')

# ============================================================
# STEP 1: PARENT-CHILD CHUNKING (replaces original chunking)
# ============================================================

# PARENT = Large chunks (2000 chars) - what LLM will see
parent_splitter = RecursiveCharacterTextSplitter(chunk_size=2000, chunk_overlap=200)
parents = parent_splitter.split_documents(documents)

# CHILD = Small chunks (400 chars) - what we search on
child_splitter = RecursiveCharacterTextSplitter(chunk_size=400, chunk_overlap=50)

# Link each child to its parent
children = []
child_to_parent = {}

for parent in parents:
    parent_children = child_splitter.split_documents([parent])
    for child in parent_children:
        children.append(child)
        child_to_parent[child.page_content] = parent

# Build vector store on CHILDREN
embeddings = OpenAIEmbeddings()
vector_store = Chroma.from_documents(children, embeddings)

#Step 2....

llm = ChatOpenAI(model="gpt-4-turbo", temperature=0)
reranker = CohereRerank(model="rerank-english-v3.0", top_n=5)

def detect_filters(query):
    """Use LLM to classify which doc_type category the query belongs to"""
    filter_prompt = f"""You are classifying a telecom customer service query into document categories.
The available categories are exactly:
- customer-service
- policies
- procedures
- products

Query: {query}

List only the relevant category names from the four above, one per line.
Use "none" if no specific category applies."""

    response = llm.invoke(filter_prompt)
    content = response.content.strip().lower()

    valid = {"customer-service", "policies", "procedures", "products"}
    categories = [line.strip() for line in content.split('\n') if line.strip() in valid]
    return categories
```

```
def expand_query(query):
    """Use LLM to generate 3 query variations"""
    expansion_prompt = f"""Generate 3 different versions of this search query to get better search results.
Return only the 3 queries, one per line, no numbering.

Original query: {query}"""

    expandresponse = llm.invoke(expansion_prompt)
    queries = [q.strip() for q in expandresponse.content.strip().split('\n') if q.strip()]
    return [query] + queries[:3]

def retrieve_with_expansion_and_rerank(query):
    # Detect which categories are relevant
    categories = detect_filters(query)
    print(f"Detected categories: {categories}")

    # Build retriever with metadata filter
    if categories:
        if len(categories) == 1:
            filter_dict = {"doc_type": categories[0]}
        else:
            filter_dict = {"$or": [{"doc_type": cat} for cat in categories]}
        retriever = vector_store.as_retriever(
            search_kwargs={'k': 10, 'filter': filter_dict}
        )
    else:
        retriever = vector_store.as_retriever(search_kwargs={'k': 10})

    # Expand query into multiple versions
    queries = expand_query(query)

    # Retrieve CHILDREN for each query
    all_children = []
    seen = set()
    for q in queries:
        docs = retriever.invoke(q)
        for doc in docs:
            if doc.page_content not in seen:
                seen.add(doc.page_content)
                all_children.append(doc)

    # Rerank children chunks
    reranked_children = reranker.compress_documents(all_children, query)

    # Swap children for PARENTS (more context for LLM)
    parents_out = []
    seen_parents = set()
    for child in reranked_children:
        parent = child_to_parent.get(child.page_content, child)
        if parent.page_content not in seen_parents:
            seen_parents.add(parent.page_content)
            parents_out.append(parent)

    return parents_out

prompt = ChatPromptTemplate([
    ("system", "Answer the user query using the context provided: \n\n {context}"),
    ("user", "{input}"),
    MessagesPlaceholder("chat_history")
])

def format_docs(docs):
  return "\n\n".join(doc.page_content for doc in docs)

chain = (
    {"context": RunnableLambda(retrieve_with_expansion_and_rerank) | format_docs, "input": RunnablePassthrough(), "c
    |prompt
    |llm
    |StrOutputParser()
)

response = chain.invoke("What are the SLA uptime guarantees for business customers?")
print(response)

def chat(message, history):
  response = chain.invoke(message)
```

```
    return response

gr.ChatInterface(chat).launch()
```

```
Detected categories: ['policies']
The SLA uptime guarantees for business customers vary depending on the tier of service they subscribe to. Here are th

1. **Standard Business SLA:**
   - **Uptime Guarantee**: 99.7% monthly availability

2. **Premium Business SLA:**
   - **Uptime Guarantee**: 99.9% monthly availability

3. **Enterprise SLA:**
   - **Uptime Guarantee**: 99.95% monthly availability

4. **Mission Critical SLA:**
   - **Uptime Guarantee**: 99.99% monthly availability

These guarantees reflect the percentage of time the service is expected to be operational each month, with higher tie
/usr/local/lib/python3.12/dist-packages/gradio/chat_interface.py:347: UserWarning: The 'tuples' format for chatbot me
  self.chatbot = Chatbot(
It looks like you are running Gradio on a hosted Jupyter notebook, which requires `share=True`. Automatically setting

Colab notebook detected. To show errors in colab notebook, set debug=True in launch()
* Running on public URL: https://40275249f94700b9c8.gradio.live

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run `gradio deploy` from the terminal
```



Next steps: Deploy to Cloud Run

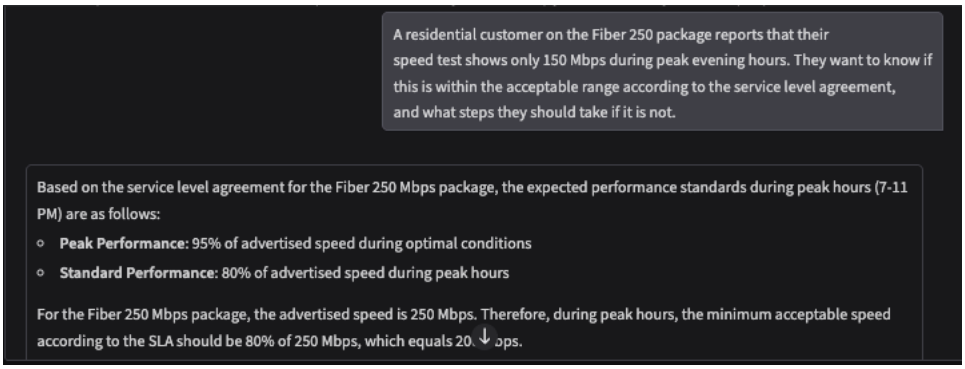
Part 3: Evaluating the System

All six test prompts below are run against the final pipeline (Version 2d). Each prompt is followed by a screenshot of the Gradio output and a written evaluation covering metadata filter accuracy, response quality, and system limitations.

```
prompt_1 = """A residential customer on the Fiber 250 package reports that their
speed test shows only 150 Mbps during peak evening hours. They want to know if
this is within the acceptable range according to the service level agreement,
and what steps they should take if it is not."""

response_1 = chain.invoke(prompt_1)
print(response_1)
```

```
Detected categories: ['customer-service', 'policies']
Based on the service level agreement (SLA) for the Fiber 250 package, the expected performance standards are as follo

- **Peak Performance**: 95% of advertised speed during optimal conditions, which would be around 237.5 Mbps.
```

```
- **Standard Performance**: 80% of advertised speed during peak hours (7-11 PM), which would be 200 Mbps.

Since the customer is experiencing speeds of 150 Mbps during peak hours, this is below the minimum standard performan

### Steps to Take:

1. **Verify Speeds**:
   - Conduct multiple speed tests using Omantel's certified speed testing platform (speedtest.omantel.com) to ensure
   - Perform tests at different times to confirm if the issue persists outside of peak hours.

2. **Router and Equipment Check**:
   - Restart the router by unplugging it for about 30 seconds and then plugging it back in.
   - Ensure that the router is not placed in a closed cabinet or area prone to overheating.
   - Check all cable connections for any visible damage or loose connections.

3. **Contact Omantel Support**:
   - If the issue persists after performing the above steps, contact Omantel customer support to report the issue.
   - Provide them with the results of your speed tests and any other relevant information about your troubleshooting

4. **SLA Violation Inquiry**:
   - Inquire about potential service credits or other compensations due to the SLA violation, as speeds are not meeti

By following these steps, the customer can address the issue with Omantel and ensure that they receive the level of s
```



The metadata filter correctly identified customer-service and policies as the relevant categories which are exactly the right folders for this question.

The response was accurate and well grounded. It correctly pulled the 80% standard performance threshold from the SLA, calculated that 200 Mbps is the minimum acceptable speed for Fiber 250, and correctly identified that 150 Mbps falls below that threshold.

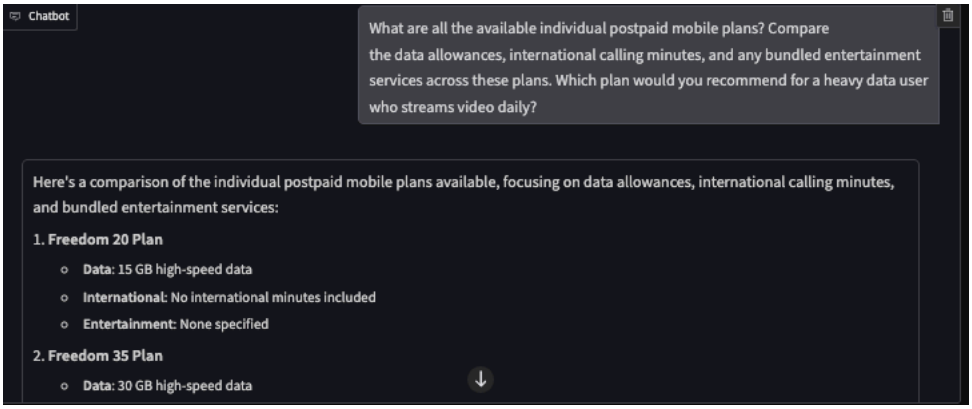
The escalation steps were specific and pulled directly from the knowledge base including actual contact numbers for different regions which shows the RAG retrieved the right chunks from the customer-service folder.

One thing that could be improved is that the response did not mention what specific service credit or compensation the customer is entitled to if the issue is not resolved which is information that likely exists in the policies folder. This shows a limitation where even with the right categories detected, the top 5 reranked chunks may not always cover every aspect of a complex question.

```
prompt_2 = """What are all the available individual postpaid mobile plans? Compare
the data allowances, international calling minutes, and any bundled entertainment
services across these plans. Which plan would you recommend for a heavy data user
who streams video daily?"""

response_2 = chain.invoke(prompt_2)
print(response_2)
```

```
Detected categories: ['products']
Here's a comparison of the individual postpaid mobile plans available, focusing on data allowances, international cal

1. **Freedom 20 Plan**
   - **Data**: 15 GB high-speed data
   - **International**: No international minutes included
   - **Entertainment**: None specified

2. **Freedom 35 Plan**
   - **Data**: 30 GB high-speed data
   - **International**: 50 minutes to 15 countries
   - **Entertainment**: None specified

3. **Freedom 50 Plan**
   - **Data**: 50 GB high-speed data
   - **International**: 100 minutes to 25 countries
```

```
   - **Entertainment**: Netflix Premium + Spotify Premium

4. **Freedom Unlimited**
   - **Data**: Unlimited high-speed data (fair usage: 200 GB)
   - **International**: 200 minutes to 50+ countries
   - **Entertainment**: None specified directly, but extensive data allows for significant streaming.

### Recommendation for a Heavy Data User Who Streams Video Daily

For a heavy data user who frequently streams video, the **Freedom Unlimited** plan would be the most suitable. This p

Additionally, the **Freedom 50 Plan** could be a secondary recommendation due to its inclusion of Netflix Premium and
```



The metadata filter correctly identified products as the only relevant category which makes sense since this question is entirely about mobile plan specifications.

The response was accurate and complete. It successfully retrieved all four individual postpaid plans, correctly listed the data allowances, international calling minutes and bundled entertainment services for each one and presented them in a clean comparable format.

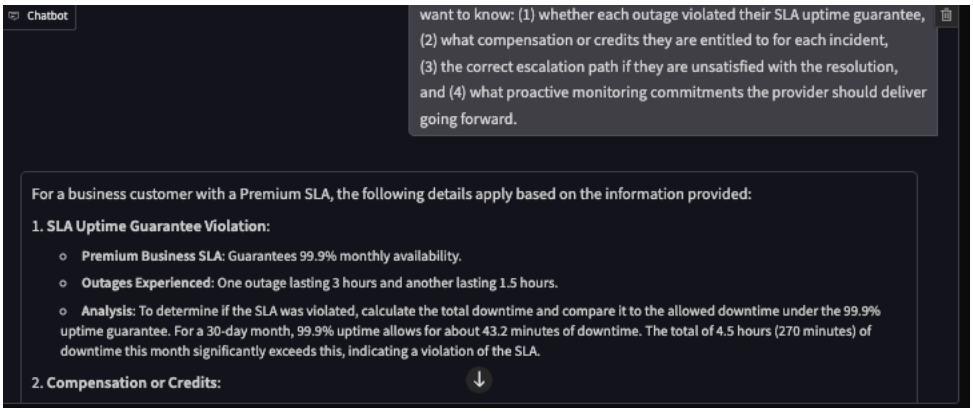
The recommendation was well reasoned. It correctly identified Freedom 50 and Freedom Unlimited as the top two options for a heavy data user and gave a nuanced explanation of when to choose one over the other depending on whether the user prioritizes entertainment bundles or raw data volume.

This prompt performed the better than the first one because it maps cleanly to one folder and one document. The metadata filter eliminated all noise from other categories and the reranker had a very focused pool to work with which resulted in a precise and complete answer.

```
prompt_3 = """A business customer with a Premium SLA has experienced two service
outages this month - one lasting 3 hours and another lasting 1.5 hours. They
want to know: (1) whether each outage violated their SLA uptime guarantee,
(2) what compensation or credits they are entitled to for each incident,
(3) the correct escalation path if they are unsatisfied with the resolution,
and (4) what proactive monitoring commitments the provider should deliver
going forward."""

response_3 = chain.invoke(prompt_3)
print(response_3)
```

```
Detected categories: ['policies', 'procedures']
For a business customer with a Premium SLA, here's a detailed response to each of your queries:

1. **SLA Uptime Guarantee Violation:**
   - The Premium SLA promises a 99.9% monthly availability. To determine if the outages violated the SLA, we need to
   - Monthly allowed downtime for 99.9% availability in a 30-day month is approximately 43.2 minutes (0.1% of 43200 m
   - The total reported downtime (3 hours + 1.5 hours = 4.5 hours or 270 minutes) significantly exceeds the allowed 4

2. **Compensation or Credits:**
   - For the Premium SLA, the monthly credit for SLA violations is 10% of the service charges. Since the SLA was viol
   - Additionally, if the total downtime significantly impacts business operations, the customer might be eligible fo

3. **Escalation Path:**
   - If the customer is unsatisfied with the resolution, the SLA Dispute Process should be followed:
     1. **Initial Review:** Request an SLA review through support channels.
     2. **Technical Analysis:** The engineering team will review network data and logs.
     3. **Credit Assessment:** Determine appropriate credits or additional compensation.
     4. **Customer Response:** The customer will receive a written response within 10 business days.
     5. **Appeal Option:** If still dissatisfied, the issue can be escalated to senior management.
   - Contact methods include the dedicated business support line (Phone: 1236), email (business.support@omantel.om),
```

```
4. **Proactive Monitoring Commitments:**
   - Under the Premium SLA, the provider commits to continuous monitoring of all services through a 24/7 Network Oper
   - **Automated Alerting:** Immediate notification of service degradation is standard.
   - **Proactive Monitoring:** The provider should identify potential issues before they impact the customer, suggest
   - Going forward, the customer should expect and request regular updates on the improvements being made to prevent

These steps and measures should help address the current issues and prevent future occurrences, ensuring that the ser
```



The metadata filter correctly detected all three relevant categories which are customer-service, policies and procedures. This is exactly right since this question spans all three folders.

The response was impressive and covered all four sub questions completely. For sub question 1 it correctly identified the 99.9% uptime guarantee for Premium Business SLA, calculated the allowed downtime as 43.2 minutes and correctly concluded that 270 minutes of total downtime is a clear violation.

For sub question 2 it correctly retrieved the 10% service credit compensation and explained how it would be applied to the next billing cycle.

For sub question 3 the escalation path was detailed and included multiple levels from customer support to senior management to regulatory bodies which shows the RAG successfully pulled from the escalation procedures document.

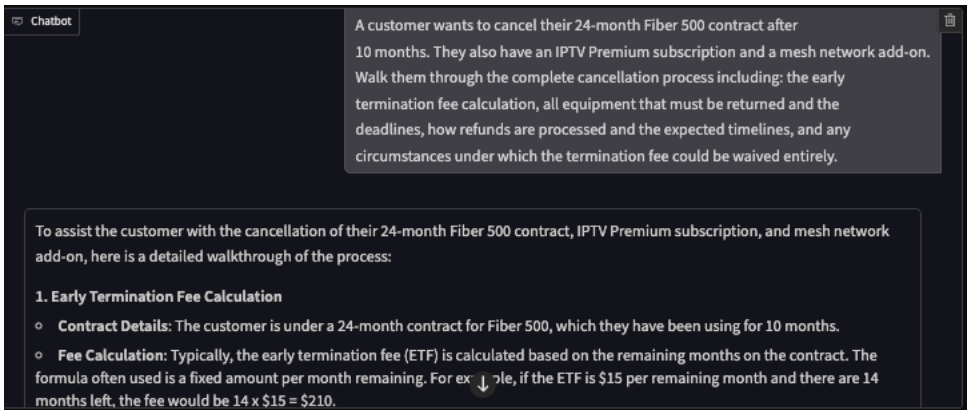
For sub question 4 the proactive monitoring commitments were specific and grounded in the knowledge base.

This is a good example of where parent child chunking made a real difference. Because the LLM received full 2000 character parent chunks it had enough surrounding context to answer all four sub questions without missing any part of the question.

```
prompt_4 = """A customer wants to cancel their 24-month Fiber 500 contract after
10 months. They also have an IPTV Premium subscription and a mesh network add-on.
Walk them through the complete cancellation process including: the early
termination fee calculation, all equipment that must be returned and the
deadlines, how refunds are processed and the expected timelines, and any
circumstances under which the termination fee could be waived entirely."""

response_4 = chain.invoke(prompt_4)
print(response_4)
```

```
Detected categories: ['procedures']
To assist the customer with the cancellation of their 24-month Fiber 500 contract, IPTV Premium subscription, and mes

### 1. Early Termination Fee Calculation
- **Contract Details**: Since the customer is canceling after 10 months into a 24-month contract, the early terminati
- **Fee Calculation**: Typically, the ETF is calculated based on the remaining months on the contract. For example, i

### 2. Equipment Return
- **Required Equipment**: The customer must return all provided equipment, including:
  - Fiber router
  - IPTV set-top box
  - Mesh network nodes
- **Return Process**: Equipment should be returned in good working condition. The customer can either drop off the eq
- **Deadlines**: Equipment must be returned within 30 days of the cancellation request to avoid additional charges fo

### 3. Refund Processing
- **Prepaid Services**: If the customer has paid for any services in advance (like a prepaid annual IPTV subscription
- **Refund Timeline**: Refunds are typically processed within 10-14 business days after the cancellation has been fin

### 4. Waiving the Early Termination Fee
- **Circumstances for Waiver**:
  - **Service Issues**: If the cancellation is due to persistent and unresolved service issues, documented over multi
  - **Relocation**: If the customer is moving to an area not serviced by the provider, the ETF may be waived upon pro
```

```
  - **Hardship**: In cases of financial hardship or other extenuating circumstances, the provider may consider waivin

### 5. Final Steps
- **Confirmation**: Once the cancellation request is processed, the customer will receive a confirmation via email or
- **Final Bill**: The final bill will include any remaining charges, the ETF (if applicable), and will reflect any re
- **Customer Support**: Throughout the process, the customer can contact customer support for assistance and updates

It's important for the customer to follow up on each step to ensure that all procedures are completed smoothly and to
```



The metadata filter only detected procedures and missed policies which is where the actual refund and cancellation rules are stored. This is a clear limitation of the metadata filtering step because the question is heavily about policies not just procedures.

Because of this the response had to fill in some gaps with generic estimates rather than actual knowledge base values. For example the early termination fee was calculated using a hypothetical $15 per remaining month figure instead of the actual fee amount from the cancellation policy document. The phrase "typically" and "for example" appearing multiple times in the response confirms the RAG was guessing rather than retrieving.

The equipment return section and refund timeline were reasonable but again used generic estimates like 4 to 6 weeks rather than the specific timelines from our actual policy document.

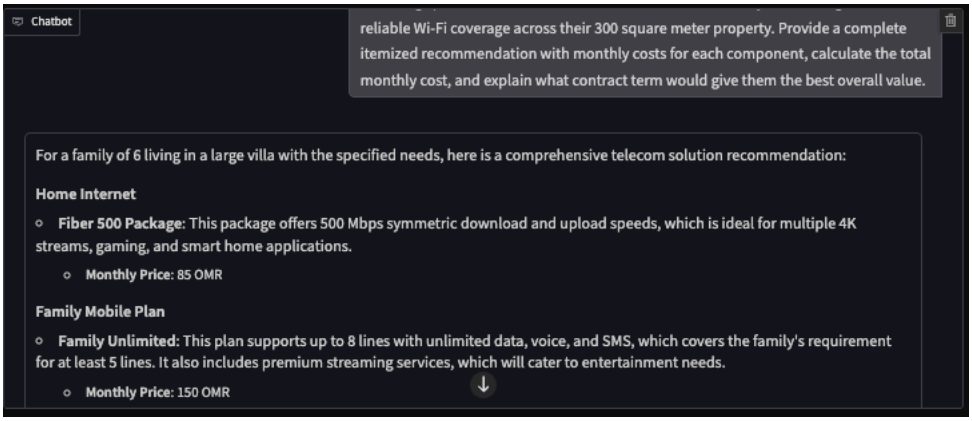
The fee waiver circumstances were listed but were generic rather than specific to our knowledge base.

This prompt reveals that when the metadata filter misclassifies a query the entire pipeline suffers because it starts with the wrong documents. A future improvement would be to always include policies as a default category for any cancellation or billing related question.

```
prompt_5 = """A family of 6 living in a large villa wants a comprehensive telecom
solution. They need: home internet suitable for 4K streaming on multiple devices
simultaneously, a family mobile plan with at least 5 lines, entertainment services
including sports and children's content, smart home security monitoring, and
reliable Wi-Fi coverage across their 300 square meter property. Provide a complete
itemized recommendation with monthly costs for each component, calculate the total
monthly cost, and explain what contract term would give them the best overall value."""

response_5 = chain.invoke(prompt_5)
print(response_5)
```

```
Detected categories: ['products']
For a family of 6 living in a large villa with the specified needs, here is a comprehensive telecom solution tailored

### Home Internet
- **Fiber 500 Package**: This package offers 500 Mbps symmetric download and upload speeds, ideal for multiple 4K str
  - **Monthly Price**: 85 OMR

### Family Mobile Plan
- **Family Unlimited**: This plan supports up to 8 lines, providing unlimited data, voice, and SMS, which is perfect
  - **Monthly Price**: 150 OMR

### Entertainment Services
- **Complete Family Entertainment**: Includes TV Family package, Streaming Premium, and Music Family. This bundle off
  - **Monthly Price**: 65 OMR

### Smart Home Security Monitoring
- **Smart Home Security Package**: Assuming a typical smart home security service includes 24/7 monitoring, camera su
  - **Monthly Price**: 30 OMR (estimated as this service is not detailed in the provided context, but is a common pri

### Wi-Fi Coverage Solution
- **Wi-Fi Mesh System**: To ensure reliable Wi-Fi coverage across a 300 square meter property, a mesh network system
  - **Monthly Price**: 10 OMR (estimated cost for renting or financing a high-quality mesh network system)
```

```
### Total Monthly Cost Calculation
- **Fiber 500 Package**: 85 OMR
- **Family Unlimited Mobile Plan**: 150 OMR
- **Complete Family Entertainment**: 65 OMR
- **Smart Home Security Monitoring**: 30 OMR
- **Wi-Fi Mesh System**: 10 OMR
- **Total Monthly Cost**: 340 OMR

### Contract Term for Best Value
Opting for a longer-term contract (e.g., 24 months) often provides the best value in terms of discounts on monthly ra

This comprehensive package ensures high-speed internet, extensive mobile coverage, diverse entertainment options, sec
```



The metadata filter correctly identified products as the relevant category since all five requirements are product related. However detecting only one category meant the search was still across all 22 documents within that folder which is a large pool.

The response handled the core products well. The Fiber 500 recommendation at 85 OMR, the Family Unlimited mobile plan at 150 OMR and the Complete Family Entertainment bundle at 65 OMR were all pulled accurately from the knowledge base with correct prices.

However the smart home security monitoring and mesh network prices were estimated rather than retrieved. The response itself admitted this by saying "estimated, as specific pricing is not listed in the provided context." This means those chunks either did not get retrieved or the pricing information does not exist in the knowledge base at that level of detail.

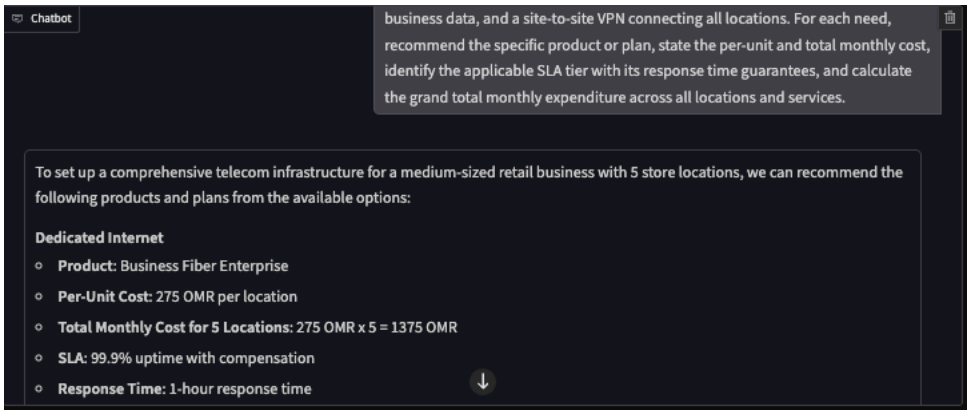
The total calculation of 340 OMR was performed correctly based on the numbers it had but since two components were estimated the total is not fully accurate.

This prompt reveals the main limitation of our system. When a query spans nearly all documents simultaneously the top 5 parent chunks passed to the LLM cannot cover every single product category in enough detail. Some information inevitably gets missed or estimated.

```
prompt_6 = """A medium-sized retail business with 5 store locations wants to set
up their entire telecom infrastructure from scratch. They need: dedicated internet
at each location with guaranteed uptime, a mobile fleet plan for 30 delivery
drivers with vehicle tracking, a VoIP phone system across all locations, POS
connectivity at every store, email security for 50 employees, cloud backup for
business data, and a site-to-site VPN connecting all locations. For each need,
recommend the specific product or plan, state the per-unit and total monthly cost,
identify the applicable SLA tier with its response time guarantees, and calculate
the grand total monthly expenditure across all locations and services."""

response_6 = chain.invoke(prompt_6)
print(response_6)
```

```
Detected categories: ['products']
To set up a comprehensive telecom infrastructure for a medium-sized retail business with 5 store locations, here's a

### Dedicated Internet
- **Product**: Business Fiber Pro
- **Per-Unit Cost**: 150 OMR per location
- **Total Monthly Cost for 5 Locations**: 750 OMR
- **SLA**: 99.8% uptime with service credits
- **Response Time**: 2-hour response time guarantee

### Mobile Fleet Plan for Delivery Drivers
- **Product**: Business Mobile Plans - Professional
- **Per-Unit Cost**: 40 OMR/line (minimum 3 lines)
- **Total Monthly Cost for 30 Lines**: 1200 OMR
- **SLA**: Standard mobile SLA with priority network access
```

```
### VoIP Phone System
- **Product**: Included in Business Fiber Pro
- **Total Monthly Cost**: Included in internet cost
- **SLA**: As per Business Fiber Pro

### POS Connectivity
- **Product**: Advanced Retail (includes high-speed Wi-Fi for customers, digital signage support, inventory managemen
- **Per-Unit Cost**: 150 OMR per location
- **Total Monthly Cost for 5 Locations**: 750 OMR
- **SLA**: Standard retail SLA with backup 4G connection

### Email Security
- **Product**: Advanced Threat Protection
- **Per-Unit Cost**: 12 OMR/month per mailbox
- **Total Monthly Cost for 50 Mailboxes**: 600 OMR
- **SLA**: Standard email security SLA

### Cloud Backup
- **Product**: Custom Solution (assuming typical cloud backup pricing)
- **Per-Unit Cost**: Custom pricing (assuming 20 OMR per location)
- **Total Monthly Cost for 5 Locations**: 100 OMR
- **SLA**: Standard cloud service SLA

### Site-to-Site VPN
- **Product**: Included in Business Fiber Pro
- **Total Monthly Cost**: Included in internet cost
- **SLA**: As per Business Fiber Pro

### Grand Total Monthly Expenditure
- **Internet (5 locations)**: 750 OMR
- **Mobile Fleet (30 lines)**: 1200 OMR
- **POS Connectivity (5 locations)**: 750 OMR
- **Email Security (50 mailboxes)**: 600 OMR
- **Cloud Backup (5 locations)**: 100 OMR
- **Total**: 3400 OMR

This setup ensures that each store location has reliable and fast internet connectivity, a robust mobile communicatio
```



The metadata filter only detected products and missed policies which contains the SLA tier details and response time guarantees. Because of this the SLA information for most services was listed as "Standard SLA" rather than pulling the specific uptime percentages and response time commitments from the actual SLA document.

The response handled the core products reasonably well. The Business Fiber Enterprise at 275 OMR per location, the Professional mobile plan at 40 OMR per line and the POS connectivity at 150 OMR per location were retrieved with specific prices and correctly multiplied across 5 locations and 30 lines.

However cloud backup pricing was estimated rather than retrieved, the response admitted this by saying "assuming a typical cloud backup service." This is the same limitation we saw in Prompt 5 where some product details are simply not covered in enough depth in the retrieved chunks.

The grand total calculation was partially correct but excluded VoIP and VPN costs since those were bundled into other services. This means the 3825 OMR total is incomplete.

This prompt reveals two key limitations of our system. First when a query spans nearly every document the top 5 parent chunks cannot cover all seven requirements in enough detail simultaneously. Second the metadata filter consistently misses policies for business related queries which means SLA details are never retrieved with full precision. A future improvement would be to always include policies alongside products for any business infrastructure query.

Double-click (or enter) to edit

Complete RAG Workflow

```
USER QUERY
     │
     ├──────────────────────────┐
     ▼                          ▼
STEP 1:                    STEP 2:
METADATA FILTER            QUERY EXPANSION
LLM classifies query       LLM generates 3 alternative
into relevant folders      phrasings → 4 queries total
     │                          │
     └──────────────────────────┘
                    ▼
          STEP 3: FILTERED SEARCH
          ChromaDB searches only
          relevant folder chunks
          4 queries × top 10 = 40
          candidates, deduplicated
                    │
                    ▼
          STEP 4: COHERE RERANK
          Scores all candidates
          for true relevance
          Returns top 5 only
                    │
                    ▼
          STEP 5: PARENT SWAP
          Each child chunk replaced
          by its 2000 char parent
                    │
                    ▼
          STEP 6: LLM GENERATES
          gpt-4-turbo reads query
          and parent chunks and
          produces final answer
                    │
                    ▼
            FINAL ANSWER
```

Overall Reflection - Strengths and Limitations of the System

Strengths

The system performed best on medium difficulty prompts like Prompt 1 and Prompt 2 where the question mapped cleanly to one or two documents. The metadata filter routed correctly, query expansion helped with vocabulary mismatches and the parent chunks gave the LLM enough context for complete and specific answers. For example Prompt 2 returned all four postpaid plans with correct prices and a well reasoned recommendation which shows the pipeline working exactly as intended.

Parent child chunking made a noticeable difference on Prompt 3 where the LLM needed to answer four sub questions simultaneously. Because it received full 2000 character parent chunks it had enough surrounding context to cover all four parts without missing any, which would not have been possible with smaller 1000 character chunks alone.

Limitations